LLL degeneracy probes, re-run 2026-10-16

This report records observations only. It re-runs the committed probes behind `papers/lll-degeneracy-note` to correct that note (sheriff case SHF-20261016-3caa6f, blocker P1). Code: the commit in `logs/commit.txt`. Commands and logs are in `README.md` and `logs/`.

Correction (2026-10-17, sheriff case SHF-20261017-e4d7f6, P-N1 and P-N2). Counts were probe runs of deterministic, overlapping tests; distinct seed pairs are now given. Every P-521 probe ran at $k_{\text{bits}} = 384$, not 390 (= $\lfloor 3 \cdot 521/4 \rfloor$). The attribution of the P-521 scaled-`f64` failure to catastrophic cancellation is withdrawn as unsupported (with the scaling, P-521's diagonal is exactly 2^{500} in `f64`); the measured results stand. The figure counts runs; every share is 0 or 1 either way.

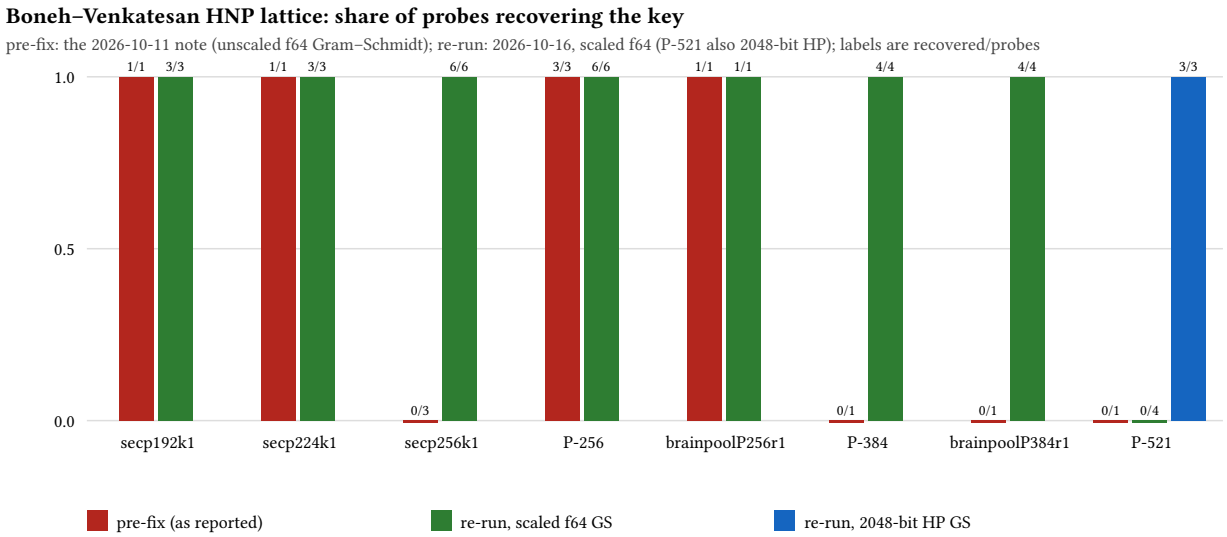

Figure 1: Share of probes recovering the planted key: the note's pre-fix reports (2026-10-11) against the 2026-10-16 re-run (`results.json`).

- **secp256k1:** 3/3 distinct seed pairs recovered (6/6 probe runs). The note reported 0/3 and called it "curve-specific".
- **P-384 and brainpoolP384r1:** 3/3 distinct seed pairs each (4/4 runs). The note reported iteration-cap failures.
- **P-521** ($k_{\text{bits}} = 384$, not 390): 0/3 distinct seed pairs (0/4 runs) on the scaled `f64` path, and 3/3 with 2048-bit fixed-point Gram–Schmidt.
- **Every 256- and 384-bit probe** finished in 13–27 ms (one unisolated 4-vCPU VM; not a benchmark).

Both failure modes were the `f64` Gram–Schmidt bugs that `research/notes/lattice-hnp/RESEARCH_LLL_GS_ANALYSIS.md` §10 recorded: an overflow of n^4 (secp256k1, all 384-bit curves) ~~and catastrophic cancellation (P-521)~~. P-521 fails on the scaled `f64` path for a reason not yet isolated.

Limits: 1 to 3 distinct seed pairs (1 to 6 probe runs) per curve at $m = 8$. These results show the reported failures do not reproduce. They do not make a security claim or a cost claim.